# Sets, Relations and Functions

From the class book: Chapter 1, Sections 1.1 to 1.4

## 1. Key definitions

**Power set.** $P(A)$ is the set of all subsets of $A$. If $|A| = n$ then $|P(A)| = 2^n$.

**Symmetric difference.** $A \triangle B = (A - B) \cup (B - A)$. The pair $(P(X), \triangle)$ is an abelian group with identity $\emptyset$, and every element is its own inverse.

**Relation.** A relation from $A$ to $B$ is any subset of $A \times B$; a relation on $A$ is a subset of $A \times A$.

A relation $R$ on $A$ is called

- **reflexive** if $(a, a) \in R$ for every $a \in A$; **irreflexive** if $(a, a) \notin R$ for every $a$;
- **symmetric** if $(a, b) \in R \Rightarrow (b, a) \in R$;
- **antisymmetric** if $(a, b), (b, a) \in R \Rightarrow a = b$; **asymmetric** if $(a, b) \in R \Rightarrow (b, a) \notin R$;
- **transitive** if $(a, b), (b, c) \in R \Rightarrow (a, c) \in R$.

**Equivalence relation.** A relation that is reflexive, symmetric and transitive. The equivalence class of $a$ is $[a] = \{x : x \sim a\}$.

**Partial order.** A relation that is reflexive, antisymmetric and transitive (for example $\leq$ on $\mathbb{R}$, divisibility on $\mathbb{N}$).

**Partition.** A family of non-empty, pairwise disjoint subsets of $A$ whose union is $A$.

**Function.** $f : A \to B$ assigns to each $a \in A$ exactly one $f(a) \in B$. It is *one-one* (injective) if $f(x) = f(y) \Rightarrow x = y$, *onto* (surjective) if $f(A) = B$, and a *bijection* if it is both.

## 2. Main results

### Set identities that are often tested

- $P(A) \cap P(B) = P(A \cap B)$, but in general only $P(A) \cup P(B) \subseteq P(A \cup B)$.
- $A - B = B - A \iff A = B$; $A \cup B = A \cap B \iff A = B$; $A - (A - B) = A \cap B$.
- $(X \times Y) \cap (Y \times X) = (X \cap Y) \times (X \cap Y)$, which has $|X \cap Y|^2$ elements.
- Inclusion and exclusion: $|A \cup B \cup C| = \sum |A| - \sum |A \cap B| + |A \cap B \cap C|$.

**Theorem (Equivalence relations and partitions)**

The equivalence classes of an equivalence relation on $A$ form a partition of $A$, and conversely every partition of $A$ comes from exactly one equivalence relation ("same block"). Two classes are either equal or disjoint, and $[a] = [b] \iff a \sim b$.

Consequently, the number of equivalence relations on an $n$-set equals the number of partitions, the **Bell number** $B_n$; those with exactly $k$ classes number $S(n, k)$, the **Stirling number of the second kind**.

**Bell and Stirling numbers**

$$B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_k, \qquad B_n = \sum_{k=1}^{n} S(n, k), \qquad S(n, 2) = 2^{n-1} - 1, \qquad S(n, n-1) = \binom{n}{2}.$$

| $n$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| $B_n$ | 1 | 1 | 2 | 5 | 15 | 52 | 203 | 877 |

Bell triangle: begin each row with the last entry of the previous row; each further entry is its left neighbour plus the entry directly above that neighbour. Rows: 1; 1, 2; 2, 3, 5; 5, 7, 10, 15; … The first column reads 1, 1, 2, 5, 15, 52, …

**Proposition (Composition)**
Let $f : A \to B$ and $g : B \to C$.

- $g \circ f$ one-one $\Rightarrow f$ one-one; $g \circ f$ onto $\Rightarrow g$ onto.
- A composition of bijections is a bijection, and $f$ is a bijection iff it has an inverse.
- If $A$ is finite, a map $A \to A$ is one-one $\iff$ it is onto.

Proof idea: if $f(x) = f(y)$ then $g(f(x)) = g(f(y))$, so $x = y$; and every $c \in C$ equals $g(f(a))$, which lies in $g(B)$. The bijections of a set $X$ onto itself form the symmetric group $S_X$.

## 3. Counting tables

### Relations on an $n$-element set

Method: think of the relation as an $n \times n$ 0-1 matrix with $n$ diagonal cells and $\binom{n}{2}$ pairs of off-diagonal cells $\{(a,b),(b,a)\}$. Count the choices for each diagonal cell and each off-diagonal pair, then multiply.

| Type of relation | Choices (diagonal; each pair) | Number | $n = 3$ |
|---|---|---|---|
| all relations | 2 per cell | $2^{n^2}$ | 512 |
| reflexive | 1; 4 | $2^{n^2-n}$ | 64 |
| irreflexive | 1; 4 | $2^{n^2-n}$ | 64 |
| symmetric | 2; 2 | $2^{n(n+1)/2}$ | 64 |
| reflexive and symmetric | 1; 2 | $2^{n(n-1)/2}$ | 8 |
| antisymmetric | 2; 3 | $2^n\,3^{n(n-1)/2}$ | 216 |
| asymmetric | 1 (empty); 3 | $3^{n(n-1)/2}$ | 27 |
| symmetric and antisymmetric | 2; 1 (empty) | $2^n$ | 8 |
| equivalence relations | no product rule | $B_n$ | 5 |
| partial orders | no product rule | 1, 3, 19, 219 for $n = 1, \dots, 4$ | 19 |

From $A$ to $B$ with $|A| = m$, $|B| = n$ there are $2^{mn}$ relations.

### Functions $A \to B$, $|A| = m$, $|B| = n$

| Type | Number | $m = 4,\ n = 3$ |
|---|---|---|
| all functions | $n^m$ | 81 |
| one-one (needs $m \le n$) | $n(n-1)\cdots(n-m+1)$ | 0 |
| onto (needs $m \ge n$) | $n!\,S(m,n) = \sum_{k=0}^{n}(-1)^k\binom{n}{k}(n-k)^m$ | 36 |
| bijections ($m = n$) | $n!$ | none |
| binary operations on an $n$-set | $n^{n^2}$ | |
| commutative binary operations | $n^{n(n+1)/2}$ | |

## 4. Exam shortcuts and standard counter-examples

**Exam shortcut.** Memorise $B_n$: 1, 2, 5, 15, 52, 203. Equivalence relations on a 4-set: 15; on a 5-set: 52. Equivalence relations with exactly two classes on an $n$-set: $2^{n-1}-1$.

**Exam shortcut.** The empty relation on a non-empty set is symmetric, antisymmetric, asymmetric and transitive, but *not* reflexive. So "symmetric + transitive $\Rightarrow$ reflexive" is false.

**Exam shortcut.** "Symmetric" and "antisymmetric" are not opposites: the identity relation, or $\{(a,a),(b,b)\}$, is both. Every asymmetric relation is antisymmetric (and irreflexive).

**Exam shortcut.** Standard examples: on $\mathbb{Z}$, $a \sim b \iff n \mid a-b$ is an equivalence relation with $n$ classes forming $\mathbb{Z}_n$; on $\mathbb{R}$, $x \sim y \iff x-y \in \mathbb{Q}$ is an equivalence relation with uncountably many classes; $|x-y| \le 1$ is reflexive and symmetric but not transitive; $x \le y$ is a partial order, not an equivalence relation; $a < b$ is not a partial order (not reflexive).

**Exam shortcut.** The number of onto maps is *not* $\binom{m}{n}n!$; use inclusion and exclusion or $n!\,S(m,n)$. Transitivity can never be counted by a product rule: use the Bell numbers or the partial-order list.

## 5. Solved examples

**Example 1.** Find the number of relations on $\{1,2,3,4\}$ that are both reflexive and antisymmetric.

***Solution.*** Reflexive forces all 4 diagonal cells to be 1. For each of the $\binom{4}{2}=6$ off-diagonal pairs $\{(a,b),(b,a)\}$, antisymmetry allows three choices: neither, only $(a,b)$, only $(b,a)$. Hence the number is $1 \cdot 3^6 = 729$.

**Example 2.** Find the number of onto functions from a 6-element set to a 3-element set.

***Solution.*** By inclusion and exclusion,

$$3^6 - \binom{3}{1}2^6 + \binom{3}{2}1^6 - 0 = 729 - 192 + 3 = 540.$$

Check: $S(6,3)=90$ and $3! \cdot 90 = 540$.

**Example 3.** How many equivalence relations on $\{1,2,3,4,5\}$ contain the pair $(1,2)$? How many have exactly three classes?

***Solution.*** An equivalence relation containing $(1,2)$ is a partition in which 1 and 2 lie in the same block. Glue 1 and 2 into a single element; we then partition a 4-element set, so the answer is $B_4 = 15$.
Exactly three classes: $S(5,3)$. The block sizes are $3+1+1$ ($\binom{5}{3}=10$ ways) or $2+2+1$ ($\frac{1}{2}\binom{5}{2}\binom{3}{2}=15$ ways), so $S(5,3)=25$.

**Example 4.** How many relations on a 3-element set are both symmetric and antisymmetric? How many are reflexive and symmetric on a 4-element set?

***Solution.*** Symmetric and antisymmetric together forbid every off-diagonal entry (if $(a,b) \in R$ with $a \neq b$ then $(b,a) \in R$, contradicting antisymmetry). The diagonal is free: $2^3 = 8$ relations (the subsets of the identity relation).
Reflexive and symmetric on 4 points: diagonal fixed, 2 choices per off-diagonal pair, so $2^6 = 64$.

**Example 5.** Which of the following are equivalence relations? (a) on $\mathbb{R}$, $x \sim y \iff x - y \in \mathbb{Z}$; (b) on $\mathbb{R} \setminus \{0\}$, $x \sim y \iff xy > 0$; (c) on $\mathbb{R}$, $x \sim y \iff xy \geq 0$; (d) on $\mathbb{R}$, $x \sim y \iff x^2 = y^2$.

***Solution.*** (a) Yes; the classes are the cosets $x + \mathbb{Z}$, one for each point of $[0, 1)$.
(b) Yes: reflexive since $x^2 > 0$; symmetric; transitive since $x, y$ have the same sign and $y, z$ have the same sign. Two classes: positives and negatives.
(c) No: $1 \sim 0$ and $0 \sim -1$, but $1 \cdot (-1) < 0$; transitivity fails.
(d) Yes; the classes are $\{x, -x\}$. Answer: (a), (b), (d).

**Example 6.** Let $f : A \to B$, $g : B \to C$ with $g \circ f$ a bijection. Must $f$ or $g$ be a bijection?

***Solution.*** We only get $f$ one-one and $g$ onto. Counter-example: $A = C = \{1\}$, $B = \{1, 2\}$, $f(1) = 1$, $g(1) = g(2) = 1$. Then $g \circ f$ is the identity of $\{1\}$, but $f$ is not onto and $g$ is not one-one. If $A, B, C$ are finite with $|A| = |B| = |C|$, then both are bijections.

**Example 7.** Let $|A| = 5$, $|B| = 6$, $|A \cap B| = 3$. Find $|P(A) \cap P(B)|$, $|P(A) \cup P(B)|$ and $|(A \times B) \cap (B \times A)|$.

***Solution.*** $P(A) \cap P(B) = P(A \cap B)$ has $2^3 = 8$ elements. Then $|P(A) \cup P(B)| = 32 + 64 - 8 = 88$ (strictly less than $|P(A \cup B)| = 2^8 = 256$). Finally $(A \times B) \cap (B \times A) = (A \cap B)^2$ has $3^2 = 9$ elements.